

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: SIMILARITY AND ENLARGEMENT (FINAL EXPLANATION)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation of the sufuria puzzle from Mama Achieng's stall at Kibuye Market in Kisumu. Use what you measured, drew and calculated in Lessons 1 to 8. Write your answers in the space under each prompt. Use $\pi = 3.14$. Show your working for every calculation and include units (cm, cm², cm³, KSh). Where a question asks you to explain, give a reason in full sentences and use the words similar, scale factor k , area and volume. Round only at the end and say 'about'.

Part 1: Are the three sufurias similar?

Mama Achieng sells githeri in a small (S), medium (M) and large (L) sufuria. S has height 12 cm and diameter 16 cm, M has height 18 cm and diameter 24 cm, and L has height 24 cm and diameter 32 cm.

(a) Calculate the ratio of the heights and the ratio of the diameters of M to S and of L to S.

(b) Use your results to explain whether the three sufurias are similar. In your answer, say what must be true of corresponding angles and corresponding sides.

(c) In Lesson 2, rectangle A was 4 cm by 6 cm and rectangle C was 6 cm by 8 cm. Explain why C is not similar to A even though both are rectangles.

(d) What is the linear scale factor k from S to L?

Write your answer here:

Part 2: Enlargement and the scale factor

Triangle ABC has $AB = 3$ cm, $BC = 4$ cm and $AC = 5$ cm. It is enlarged from a centre O to give image $A'B'C'$ with $A'B' = 6$ cm, $B'C' = 8$ cm and $A'C' = 10$ cm.

(a) How can you find the centre O if you only have the drawing of the object and the image?

(b) Calculate the scale factor k from the sides.

(c) If $OA = 4$ cm, what is OA' ? Show how you know.

(d) Triangle ABC is now enlarged from O with $k = 0.5$. Give the three side lengths of the image and say whether the image is bigger or smaller than the object and whether it is nearer to or further from O.

(e) How is the triangle enlargement linked to the similarity S and L?

Write your answer here:

Part 3: Why paint costs rise faster than the price

The outside area of a sufuria to be painted is the side plus the base, $A = 2\pi r h + \pi r^2$.

(a) Calculate the outside area of S (radius 8 cm, height 12 cm) and of L (radius 16 cm, height 24 cm).

(b) Divide the area of L by the area of S. How does this number relate to $k = 2$?

(c) In Lesson 5 you enlarged a 3 cm by 2 cm rectangle with $k = 3$. Use it to explain why the area changes by this factor and not by k .

(d) Paint costs about KSh 1 for every 40 cm². Find the approximate paint cost for S and for L and say how many times more L costs.

Write your answer here:

Part 4: Why the large sufuria holds so much more githeri

- (a) Calculate the capacity of S and of L using $V = \pi r^2 h$. Divide the capacity of L by that of S.
- (b) In Lesson 6 you found that cubes of side 5 cm and 10 cm have volumes 125 cm³ and 1000 cm³. Use this to explain why the volume scale factor is k^3 .
- (c) One serving of githeri is 400 cm³. Find about how many servings S and L hold when filled to the brim.
- (d) The price of S is KSh 60 and the price of L is KSh 120. Find the price per serving for each. Who gets the better deal, and why?

Write your answer here:

Part 5: Final Explanation and prediction

Write your Final Explanation that answers the driving question: Why does a sufuria that is twice as tall and twice as wide hold far more than twice the githeri, and how can we predict how length, area and volume change when we enlarge an object?

In your answer:

(a) State the scale factor and say what happens to length, area and volume.

(b) Explain why each quantity changes as it does by saying how many lengths it depends on.

(c) Link your explanation to the capacity, the paint cost and the price of Mama Achieng's sufurias.

(d) A cylinder tin has radius 4 cm and height 10 cm (volume 502.4 cm³). A larger similar tin has $k = 1.5$. Predict the volume of the larger tin using a scale factor, then check it with $V = \pi r^2 h$.

(e) A photo 10 cm by 15 cm is enlarged to a poster with $k = 3$. By what factor does the area change, and what is the poster area? A different enlargement makes the area 9 times bigger; what is k ?

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Testing and describing similarity	Calculates all ratios correctly, states that corresponding angles are equal and all sides are in a constant ratio, and explains clearly why rectangle C is not similar to A.	Calculates most ratios correctly and states the constant-ratio idea, with a small gap in explaining why C is not similar.	Relies on shape looking the same or on equal angles only, and does not use a constant ratio of sides.
Scale factor and enlargement	Finds k from lengths and from OA'/OA , finds the centre O by joining	Finds k correctly and applies it to lengths, with minor errors in describing the centre	Confuses k with the difference between lengths or cannot find the image lengths

	corresponding points, and handles both $k = 2$ and $k = 0.5$ correctly.	or the reduction.	from k .
Calculating area and volume	Uses $A = 2\pi r h + \pi r^2 h$ and $V = \pi r^2 h$ with $\pi = 3.14$ accurately, gets all ratios (4 and 8), and finds servings and costs correctly with units.	Uses the formulas correctly with one or two arithmetic or unit slips.	Uses the wrong formulas or cannot connect the calculated figures to the ratios.
Explaining k , k^2 and k^3	Explains that quantities depend on one, two or three lengths, so they multiply by k , k^2 or k^3 , and uses the cube, rectangle and cylinder evidence. Works backwards from an area factor to k .	States k^2 and k^3 correctly with some evidence but a limited explanation of why.	Says area and volume simply double when lengths double.
Final Explanation of the phenomenon	Links capacity (8 times, about 48 servings against 6), paint (4 times, about KSh 80 against KSh 20) and price ($KSh\ 120 = 2 \times 60$) to the scale factor, and explains the bargain per serving.	Explains capacity and paint with the correct factors but only partly explains the price or the bargain.	Gives a general answer without the factors or without using Mama Achieng's data.

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